

Final Project

Sequential Circuit Design Automation System

1. MODEL TYPE

☒ Mealy Model
☐ Moore Model

2. FLIP-FLOP TYPE

☒ JK Flip-Flop
☐ T Flip-Flop

3. STATE TABLE INPUT

Input Variables
X

Output Variables
Z

State Table

Present State	X	Next State	Z
A	0	A	0
A	1	B	0
B	0	C	1
B	1	A	0
C	0	A	1
C	1	C	1

Clear Table Load Example

OUTPUT 1: FLIP-FLOP INPUT EQUATIONS

Flip-Flop Input Equations (Simplified)

State Variables: Q1 Q0

Flip-Flop	Input	Equation
FF for Q1	J1	$J1 = X \cdot Q0$
	K1	$K1 = X' + Q0$
FF for Q0	J0	$J0 = Q1 \cdot X$
	K0	$K0 = X$

K-Map (Example)

J1

	Q0	
Q1	0	1
0	0	X
1	0	1

(Simplified) $J1 = X \cdot Q0$

OUTPUT 2: SEQUENTIAL CIRCUIT DIAGRAM

• Circuit diagram is generated based on simplified equations.
• Click on components for more details.

Settings

GENERATE

EXPORT REPORT

About